

Ran Chen (She/Her/Hers)

Department of Statistics and Data Science, Washington University in St Louis

E-mail: ran.c@wustl.edu

Website: <http://ran-chen.com/>

Address: 1 Brookings Dr, Jolley Hall 530

RESEARCH INTERESTS Data-driven Decision-making, Statistical Machine Learning, Reinforcement Learning, High-dimensional Statistics, Optimization, Nonparametric Statistics, Revenue Management, Healthcare

ACADEMIC APPOINTMENTS **Assistant Professor** 2024 – Present
Department of Statistics and Data Science
Washington University in St Louis

Postdoc Associate 2022 – 2024
Laboratory for Information and Decision System
Massachusetts Institute of Technology (MIT)
Advisor: Martin Wainwright

Teaching Fellow 2022
Wharton Data Science Academy

Research Fellow 2022
The Wharton School
University of Pennsylvania
Advisor: Linda Zhao

EDUCATION **The Wharton School** September 2017 - August 2022
University of Pennsylvania, Philadelphia, USA
Ph.D. in Statistics and Data Science
Dissertation: *Estimation and Inference for Convex Functions and Computational Efficiency in High Dimensional Statistics* ([pdf](#))
Advisor: Tony Cai

Tsinghua University September 2013 - July 2017
Beijing, PRC
B.S. in Pure and Applied Mathematics

PAPERS

1. Cai, T.T., **Chen, R.**, Zhu, Y. (2021).
“Estimation and Inference for Minimizer and Minimum of Convex Functions: Optimality, Adaptivity, and Uncertainty Principles.” *The Annals of Statistics*, 52(1), 392-411. Available [here](#).
2. Cai, T.T., **Chen, R.**, Zhu, Y. (2021).
“Supplement Paper to Estimation and Inference for Minimizer and Minimum of Convex Functions: Optimality, Adaptivity, and Uncertainty Principles.” *Annals of Statistics*. Available [here](#).
3. Cai, J., **Chen, R.**, Wainwright, M., Zhao, L. (2023).
“Doubly High-Dimensional Contextual Bandits: An Interpretable Model with Applications to Assortment/Pricing” *Management Science* (forthcoming 2026+). Available [here](#).
4. Cai, J., **Chen, R.**, Yang, D., Zhu, W., Shen, H., Zhao, L. (2023).
“Network Regression and Supervised Centrality Estimation.” *Journal of American Statistical Association* (2026). Available [here](#).

5. Cai, J., **Chen, R.**, Huang Q., Wainwright, M., Zhao, L., Zhu W. (2024).
 “Poisson-MNL Bandit: Nearly Optimal Joint Assortment and Pricing with
 Decision-Dependent Customer Arrivals.”
under review at Operations Research
6. Deng, R., Chakraborty, B., **Chen, R.**, Tan, Y. (2025)
 “BFTS: Thompson Sampling with Bayesian Additive Regression Trees.” *ICML
 2026*

PREPRINTS

1. **Chen, R.** (2023).
 “Interplay Between Statistical Accuracy and Running Time Cost: a Framework
 and Three Cases.” *To be submitted to Operations Research. Available [here](#).*
2. **Chen, R.** (2023).
 “Optimal Estimation and Inference for Minimizer and Minimum of Multivariate
 Additive Convex Functions.” *To be submitted to Annals of Statistics. Available
[here](#).*
3. Cai, J., **Chen, R.**, Wainwright, M., Zhao, L. (2024).
 “Personalized Reinforcement Learning: with Applications to Business.” *to be
 submitted to Management Science*
4. Cai, J., **Chen, R.**, Wainwright, M., Zhao, L. (2024).
 “Personalized Reinforcement Learning for Sepsis Management in ICU.” *to be
 submitted to Operations Research*

WORKING PAPER

1. **Chen, R.**, Smetters, K. (2024). “Optimal Portfolio Choice with fat tails.” *to be
 submitted to Econometrica*
2. Luo Y., Liu H., **Chen, R.** (2024) “Convergence of Expected Suboptimality and
 Asymptotic Distribution of ADAM-Type Optimizers” *to be submitted to Annals of
 Statistics*
3. Teng, Y., **Chen, R.** (2025) “Personalized Reinforcement Learning with
 Nonparametric Kernel Embeddings” *to be submitted to Journal of American
 Statistical Association*

GRANTS

- **NSF–DMS 2515896** (09/2025–08/2028)
*Taming Sequential Decision-Making with Reinforcement Learning: Non-Stationarity,
 Heterogeneity, and Online/Offline Comparison*
 Role: Sole Principal Investigator

SELECTED TALKS

- Optimization and Statistics meets in AI age, *Department of Mathematical Sciences,
 Tsinghua University, June, 2025*
- Online Sequential Decision-Making with Reinforcement Learning: From Robotics to
 Human-centered tasks, *Department of Statistics, Tsinghua University, June 2025*
- Personalized Reinforcement Learning for Sepsis Management in ICU
 - *2025 IMS International Conference on Statistics and Data Science (ICSDS),
 Seville, Spain. Dec. 2025*
 - *INFORMS 2024, Seattle. Oct. 2024*

- Interpretable Personalized Online Reinforcement Learning with Applications in Business and Healthcare,
 - *Joint Statistical Meetings 2025, Nashville, Tennessee,* Aug., 2025
 - *Joint Conference on Statistics and Data Science in China (2025 JCSDS),* July 2025
- Network Regression and Supervised Centrality Estimation, *CMStatistics 2024, London, United Kingdom.* Dec. 2024
- Doubly High-Dimensional Contextual Bandits: An Interpretable Model for Joint Assortment and Pricing,
 - *Olin Business School, Washington University in St. Louis,* Sep. 2025
 - *The 8th International Conference on Econometrics and Statistics (EcoSta 2025), Waseda University, Tokyo, Japan* Aug. 2025
 - *IMS New Researchers Conference, Oregon State University* Aug. 2024
 - *Translational Data Workshop, Washington University in St Louis.* Apr. 2024
 - *Department of Applied Mathematics and Statistics, Johns Hopkins University.* Feb. 2024
 - *Department of Statistics, University of Wisconsin-Madison.* Feb. 2024
 - *Information Systems and Operations Management, Goizueta Business School, Emory University.* Feb. 2024
 - *Department of Technology, Operations, and Statistics, Stern School of Business, New York University.* Feb. 2024
 - *Department of Statistics, Stanford University.* Jan. 2024
 - *Department of Statistics, University of California, Davis.* Jan. 2024
 - *Department of Statistics and Data Science, Washington University in St. Louis.* Jan. 2024
 - *Department of Mathematics, Applied Mathematics, and Statistics, Case Western Reserve University.* Jan. 2024
 - *Department of Statistics, University of Washington.* Dec. 2023
 - *Department of Statistics, Harvard University.* Nov. 2023
 - *INFORMS 2023, Phoenix.* Oct. 2023
- Personalized Reinforcement Learning: with Applications to Business, *Joint Statistical Meeting 2023, Toronto.* Aug. 2023
- Dynamic joint assortment and pricing through doubly high-dimensional contextual bandits, *MSOM 2023, Montreal.* June 2023
- An Interpretable Machine Learning Model for Assortment/Pricing, *Inform Business Analytics Conference 2023, Aurora.* April 2023
- High-dimensional Continuum Armed and High-dimensional Contextual Bandit: with Applications to Assortment and Pricing, *Wharton Customer Analytics with Master Kong Food Company.* Nov. 2022
- Statistics, Optimization, and Machine Learning: with Applications in Economics and Business, *Department of Business Economic and Public Policy, The Wharton School, University of Pennsylvania.* Oct. 2022
- Crowdsourcing: Beyond Dawid Skene Model, *Joint Statistical Meeting 2020, Philadelphia.* Aug. 2020
- Heterogeneous Treatment Effect Estimation through Deep Learning, *Joint Statistical Meeting 2018, Vancouver.* Aug. 2018

SELECTED AWARDS	• The George James Doctoral Fellowship, The Wharton School	March 2017
	• XueTangBan Membership and Scholarship (Tsinghua Xuetang Mathematics Program), Tsinghua University	Feb. 2014 - July 2017
	• Academic Excellence Honor, Tsinghua University	2014, 2015, 2016
	• Tsinghua University Distinguished Student Programme (4 selected out of 107)	2014
	• Second Prize, (National) Regional College Students' Physics Contest	2014
	• Silver Medal, China Mathematical Olympiad 2013	Jan. 2013
	• Gold Medal, China Girls' Mathematical Olympiad 2012	Aug. 2012
	• Second Prize, National High Schools Physics Competition	Oct. 2012
SERVICE	• NSF Panelist.	
	• Reviewer of Annals of Statistics(1), JASA(1), JRSSB (2), The American Statistician(2), Nature Communications Medicine(2).	
	• Judge for ASA BES student paper award	
	• Member of Executive Board, Tsinghua Alumni Association of Greater Boston	Nov. 2022 - 2024
	• Director of Public Relations & Propagation and Board Director, Tsinghua Alumni Association of Greater Philadelphia	July 2021 - 2023
	• Board Member of the Wharton Society for the Advancement of Women in Business Academia	Aug. 2019 - Aug. 2021
	• Secondary Treasurer, Tsinghua Alumni Association of Greater Philadelphia	Aug. 2019 - July 2021
	• Volunteer at the 8th International Congress on Industrial and Applied Mathematics, Beijing	Aug. 2015
	• Vice President of Student Association of Science and Technology, Tsinghua Math Department	June 2015 - Dec. 2016
	• Head of Publicity, Planning, and Innovation Office of Student Association of Science and Technology, Tsinghua Math Department	June 2014 - June 2015
TEACHING	• Instructor	
	- SDS 4215/5215 Python for Data Science	Spring 2026
	- SDS 5800, Topics in Statistics - Optimization for Machine Learning	Fall 2025
	- SDS 4481, Special Topics in Statistics and Data Science: An Introduction in Python	Spring 2025
	- SDS 579, Topics in Statistics: Optimization in Machine Learning	Fall 2024
	• Teaching Fellow	
	- Wharton Data Science Academy	2022
	• TA	
	- Introduction to Python for Data Science	Spring 2022
	- Forecasting Methods for Management	Fall 2021
	- Introductory Statistics	Spring 2020, Fall 2020, Spring 2021
	- Probability	Fall 2019
	- Optimization Methods in Machine Learning	Spring 2019
	- Introduction to Business Statistics	Fall 2018

STUDENTS

- PhD student
 - Yi Luo
 - Yicheng Teng
- Undergrad student
 - Yangtong Feng
 - Haoyang Liu
 - Cindy Nan